

Phase 3: Construction — Phase Overview

Resilio — Mobile Mental Health Platform for Nepal | Milan Shrestha | ID: 23057135 | 03_Construction/README.md

RUP Phase: Construction | **Weeks:** 11–19 | **Dates:** January – March 2025

Iterations: 7 (across 9 weeks)

Phase Objective

Implement all 24 functional requirements iteratively across seven planned development iterations. Each iteration produces tested, integrated software that is demonstrated to stakeholders before the next iteration begins. Deliverables feed Chapter 3 (Development) of the final report.

Documents in This Folder

File	Contents	Report Reference
`Technology_Documentation.md`	Every Flutter package, Node.js package, and external service with version, purpose, and official documentation URL	Chapter 3, Section 3.4
`Development_Tools_and_Environment.md`	IDEs (Android Studio, VS Code), Flutter toolchain, Node.js commands, Git strategy, Postman, DBeaver, Figma, draw.io, hardware setup	Chapter 3, Section 3.4
`WebRTC_Signalling_Architecture.md`	Socket.IO event table, step-by-step signalling flow, RTCPeerConnection init code, STUN/TURN configuration, Coturn setup, session room ID design	Chapter 3, Section 3.5.5; Appendix J

File	Contents	Report Reference
`Project_Structure_and_Artefacts.md`	Annotated Flutter `lib/` tree, Node.js backend tree, SQL migration sequence, Protobuf definitions, Project Artefact folder map, week-by-week feature delivery table, code-to-report cross-reference	Chapter 3 (all sections)
`Learning_Resources.md`	YouTube channels, official documentation, tutorials, and dev.to/Medium articles used to learn every technology in the project	Reference

Development Iterations

Iteration	Weeks	Primary Features
Iter 1	11	SuperTokens OTP + Google/Facebook auth; GoRouter role guards; onboarding
Iter 2	12	Supabase schema deployed; RLS policies live; API skeleton; GetIt DI wired
Iter 3	13	Therapist directory; therapist profile management; admin verification workflow
Iter 4	14	Appointment booking; real-time slot availability; eSewa payment integration
Iter 5	15	WebRTC video consultation; Socket.IO signalling server; admin dashboard
Iter 6	16–17	Mood tracking; PHQ-9/GAD-7; crisis alert; content hub; Cloudinary; subscription gating

Iteration	Weeks	Primary Features
Iter 7	18–19	Wellness games (4 types); achievements; favourites; FCM push notifications; SQLite offline sync

Key Technical Decisions Made

Decision	Rationale
<code>`supertokens_flutter`</code> via git override	pub.dev version didn't support combined OTP + third-party recipe; master branch required
<code>`esewa_flutter_sdk`</code> as local path dep	eSewa v2 payment verification API not available on pub.dev version; SDK bundled locally
<code>`better_player_plus`</code> over <code>`video_player`</code>	Handles both long-form video and short reels (short flag) in one package
<code>`dartz`</code> Either type	Domain layer returns <code>`Either`</code> — forces callers to handle error states without exceptions
Socket.IO on separate host	Vercel serverless terminates persistent connections; Socket.IO requires persistent WebSocket host (Railway)
Protocol Buffers for offline	Binary serialisation reduces SQLite row size vs JSON; typed schema prevents corruption on sync
GetIt + Injectable	Compile-time DI registration via code generation; no runtime reflection